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Non-contact detection of electrocardiogram fiducial points with millimeter-wave radar

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ABSTRACT

Purpose: Accurate detection of fiducial points in electrocardiograms (ECG) is crucial for diagnosing heart diseases. However, traditional ECG devices require direct contact with the skin, which can cause discomfort for patients. Therefore, there is an increasing need for non-contact detection of ECG fiducial points, while research in this area remains limited.

Methods: In response to this need, we proposed a millimeter-wave radar-based method for non-contact detection of ECG fiducial points. Initially, fiducial points from synchronously acquired ECG signals were annotated to serve as labels for network training. Subsequently, a radar cardiogram (RCG) was derived from radar radio frequency signals using a second-order differentiator. To enhance the detection process, we developed a one-dimensional semantic segmentation model by integrating a deep residual shrinkage network block (DRSN) into the U-Net architecture, which we term DRSN-Unet.

Results: The results from the testing dataset demonstrate that the proposed method accurately detects the five fiducial points, achieving an average precision of 0.986, an average recall of 0.986, and an average $F_{1\text{-score}}$ of 0.986. Notably, the network's parameter count is only 2.042 Mega, with a computation amount of 0.086 GFLOPs.

Conclusion: This method effectively detects the onsets of the P-wave and QRS complex, the peak of the R-wave, and the offsets of the QRS complex and T-wave without direct contact with the human body, laying a solid foundation for the automated diagnosis of complex cardiac diseases.

1. Introduction

According to the World Health Organization, cardiovascular diseases are the most prevalent non-communicable diseases globally and a leading cause of mortality [1,2]. The electrocardiogram (ECG) serves as a fundamental tool for screening and diagnosing cardiac diseases, effectively recording and characterizing the heart's electrical activity. Traditionally, ECG monitoring is conducted using electrode sheets that contact body surface, which capture minute electrical variations during the depolarization and repolarization of myocardial tissue across multiple cardiac cycles. The ECG consists of three principal components: the P-wave, the QRS complex, and the T-wave. The P-wave is formed by atrial depolarization, the QRS complex is formed by ventricular depolarization, and the T-wave is formed by ventricular repolarization. In clinical practice, the PR, QRS, QT, and RR intervals are typically measured, with their starting and ending points of these intervals are also referred to as fiducial points. These fiducial points include the onset of the P-wave and QRS complex, the peak of the R-wave, and the offset of the QRS complex and T-wave (Fig. 1). The PR, QRS, QT, and RR intervals provide insights into the function of the cardiac conduction system and offer critical prognostic information [3, 4], aiding in diagnosing arrhythmias, including sustained premature ventricular contractions and sinus tachycardia [5]. For example, when the first-degree atrioventricular block occurs, the PR interval tends to be prolonged to more than 200 ms (ms), suggesting an increase in risk of future atrial fibrillation (approximately 2-fold) and pacemaker implantation (approximately 3-fold) [6]. Moreover, early arrhythmias can be episodic, often eluding detection during routine health check-ups, thus highlighting the need for long-term monitoring [7]. Holter monitors fulfill the requirement for continuous 24-hour monitoring, but they can cause patient discomfort and limit daily activities due to their numerous electrodes, complex wiring, and device weight.

Millimeter-wave radar, as one of the most suitable wireless radio frequency detectors for daily life, has become an emerging key technology in the healthcare and the Internet of Things industry [8]. Characterized by high frequency and resolution, millimeter-wave radar can detect micromotions at the sub-millimeter level, such as chest

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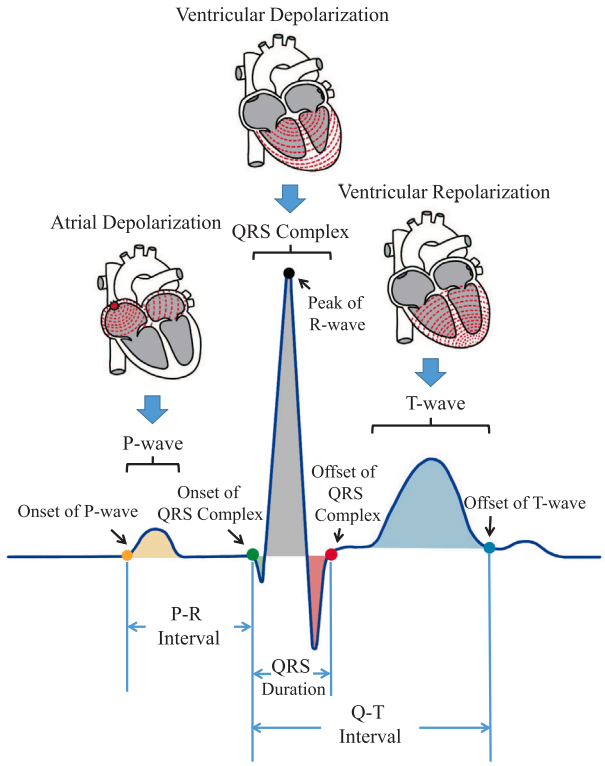

Fig. 1. Electrocardiogram fiducial points and key electrocardiogram characteristics.

wall vibrations resulting from cardiac activity. The millimeter-wave radar system emits microwaves that reflect off chest vibrations, with the reflected signals subsequently received by the antenna. A series of signal processing techniques–including range compression, target detection, and phase inversion–are employed to extract the radar phase signals from the reflected signals. This radar phase signal comprises mixed micro-motion signals from both respiration and heartbeat. Finally, signal separation techniques are utilized to isolate the heartbeat signal component from the radar phase signal [9]. Due to its non-contact characteristics, high precision, ability to penetrate clothing and thin blankets, and minimal susceptibility to environmental factors such as light, temperature, and humidity, millimeter-wave radar holds significant potential for monitoring vital signs without invasive procedures and identifying potential cardiovascular diseases [9–12].

Numerous studies have employed millimeter-wave radar for non-invasive cardiac health monitoring [13,14]. The term 'radar cardiogram (RCG)' [15] refers to the heartbeat signal component that is extracted from the radar phase signal. Studies have demonstrated a correlation between the peak positions in RCG and ECG waveforms, with their periodic fluctuations exhibiting consistency [16]. Various signal separation techniques, including wavelet transform [17], empirical wavelet transform [18], empirical modal decomposition [19], and variational mode decomposition [20], have been utilized to extract the RCG from radar phase signals. Nevertheless, these techniques frequently alter the RCG waveform and may struggle to accurately identify R-wave peaks, let alone the ECG fiducial points. Therefore, to preserve the fidelity of the separated RCG waveform relative to the ECG waveform, a second-order differentiator is employed directly to distinguish between the heartbeat and respiration signals. This approach leverages the principle that the acceleration of the heartbeat motion significantly exceeds that of the respiration signal. It computes the second-order derivative of the radar phase signal, which encompasses both the heartbeat and respiration components. The resultant signal not only mitigates the issue of RCG waveform distortion but also facilitates the extraction of additional detailed information regarding heart motion. The RCG obtained by calculating the second-order derivative shares a rhythmic pattern with the ECG, but its waveform is more complex and prone to interference from spontaneous body movements. This complexity poses a challenge when attempting to analyze the fiducial points of the ECG based on RCG data.

Recent studies have employed deep learning algorithms to establish the mapping relationship between RCG and ECG [21]. Incorporating a transformer structure into the coder–decoder network, researchers have succeeded in reconstructing the ECG from RCG with peak positioning accuracy ranging from 4 to 54 ms for Q-wave peaks, 3 to 9 ms for R-wave peaks, 3 to 40 ms for S-wave peaks, and 6 to 45 ms for T-wave peaks [22]. Another study proposed the MultiResLinkNet [23], which demonstrated that the mean RR interval, heart rates (HRs), and root mean square of consensual RR interval differences (RMSSD) calculated from the reconstructed ECG using this network approximated the actual values with minimal deviations. However, the above network structure is relatively complex and only applies to data from subjects in the lying position. A lightweight decoder–encoder network, Res-Unet, was proposed for reconstructing ECG from RCG for subjects in a sitting position [24]. This network shows good reconstruction performance for data collected from the same subject, yet its performance significantly diminishes for new subjects. Due to the variable mapping relationships between the electrical and mechanical signals of the heart across different individuals, the reconstructed ECG may not perfectly match the waveform of the original ECG. However, there exists a high degree of rhythmic consistency between RCG and ECG that can be leveraged for the rhythmic analysis of the ECG. In subsequent rhythmic analyses, it is crucial to identify fiducial points to enable further examination [25,26].

To improve the efficiency and accuracy of ECG monitoring, it is proposed that the obtained RCG be input into a deep learning network for the direct, automatic detection of ECG fiducial points. Numerous studies have investigated the automatic detection of these fiducial points [27,28]. For example, a bidirectional long short-term memory network (biLSTM) [29] has been successfully applied to detect R-wave peaks and T-wave offsets from radar heart sound signals [30], achieving a high detection accuracy of 0.934. Additionally, the U-net [31] has been applied to ECG segmentation and classification, yielding impressive classification and the peak of R-wave detection accuracy (0.973 and 0.993). A hybrid model, which integrates one dimensional (1D) convolutional layers with biLSTM, has also been introduced for segmenting the P-wave, QRS complex, and T-wave from ECG signals [32], with a notable segmentation accuracy of 0.954. Nevertheless, it is important to note that these studies were not specifically designed for RCG data. RCG waveforms differ markedly from those of traditional ECGs and exhibit greater heterogeneity. These characteristics present unique challenges for the non-contact detection of ECG fiducial points using millimeter-wave radar technology.

This study aims to leverage the principle that RCG waveforms have a mapping relationship with ECG waveforms to automatically detect five ECG fiducial points: the onset of the P-wave and QRS complex, the peak of the R-wave, the offset of the QRS complex and the T-wave. Initially, the RCG was extracted from the radar phase signal, which contained a mixture of respiratory and heartbeat signals, using an effective second-order differentiator. Subsequently, a 1D convolutional semantic segmentation network called deep residual shrinkage network block (DRSN) U-net was designed, specifically for fiducial point detection. Additionally, five existing network models were selected, including three networks previously used for ECG reconstruction (CNN-Transformer [22], MultiResLinkNet [23], Res-Unet [24]) and two classical one-dimensional data segmentation networks (CNN-biLSTM [32] and 1D-Unet [31]), to compare their effectiveness with that of our proposed model.

This paper contributes to the field in two major ways:

1. It marks the first attempt at non-contact detection of five ECG fiducial points using millimeter-wave radar and establishes a foundation for the automated diagnosis of complex cardiac diseases.

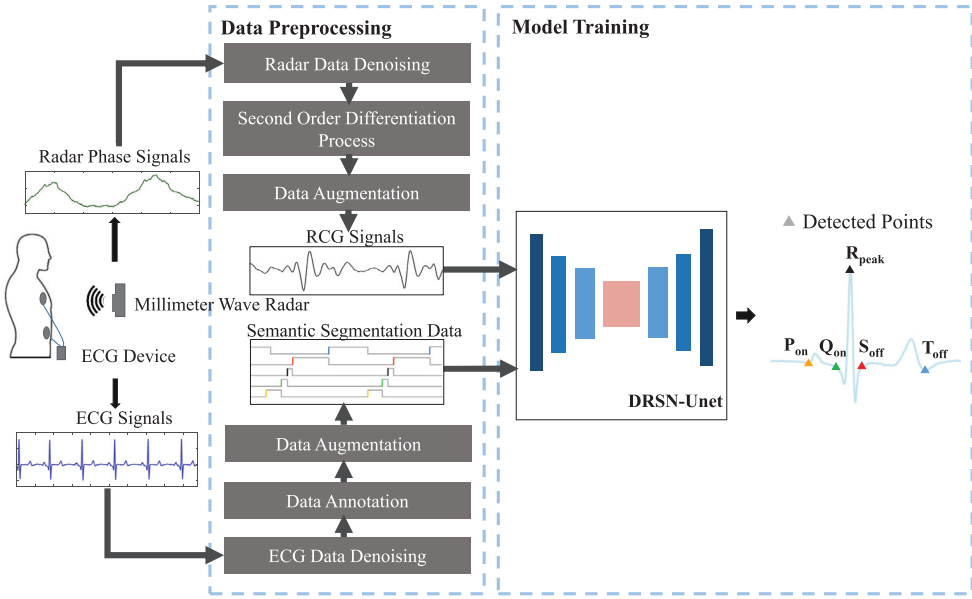

Fig. 2. The framework of our proposed algorithm.

2. The DRSN-Unet has been proposed, which is optimized for point-by-point semantic segmentation of 1D signals, capable of identifying ECG fiducial points in a non-contact manner. The experimental results showed that for the five types of fiducial points, the average $F_{1\text{-score}}$ of the model exceeded 0.986, demonstrating excellent performance.

The structure of this article is organized as follows: Section 2 describes a non-contact detection method for ECG fiducial point based on millimeter-wave radar. Section 3 provides a detailed description of the dataset and experimental setup. Section 4 presents and discusses the results of ECG fiducial point detection utilizing the proposed model alongside other deep learning methods. Finally, Section 5 summarizes the study and addresses its limitations and future research directions.

2. Methods

To assess the feasibility and effectiveness of the proposed algorithm, a publicly accessible radar dataset of clinically recorded radar vital signs, along with synchronized reference sensor signals, was obtained from Professor Sven's team at Hamburg Institute of Technology [33]. Radar signals and lead II ECG signals were synchronously collected from all 30 subjects. Additionally, radar signals and lead II ECG signals data of two subjects from another center [24] were selected to verify the effectiveness of the proposed network.

The overall framework of our proposed algorithm for non-contact ECG fiducial point detection based on millimeter-wave radar is illustrated in Fig. 2. This framework consists of two primary modules: the data preprocessing module and the network training module. In the data preprocessing module, noise interference from the synchronously collected radar phase signal and corresponding ECG signal was first eliminated. Subsequently, the RCG signal was extracted from the radar phase signal using a second-order differentiator. Automatic recognition and manual calibration of five fiducial points were then performed on the synchronously collected ECG signals to generate semantic segmentation data for these fiducial points, which serves as the label for network training. The network training module involved the design of the DRSN-Unet. Both RCG and semantic segmentation data were input simultaneously for training. Finally, the performance of the network model was evaluated on the testing dataset. Detailed descriptions of each step of the algorithm are provided below.

2.1. Data preprocessing

2.1.1. Radar data preprocessing

Firstly, the radar data was processed using a low-pass Butterworth filter with a cut-off frequency of 10 Hz to eliminate high-frequency interference. Subsequently, the RCG was extracted. Because the radar phase signal is a composite displacement resulting from both the inhale–exhale process and the pulsatile effect of the heartbeat. Since the displacements associated with the inhale–exhale process are several orders of magnitude larger than the subtle vibrations caused by heartbeat activity, the radar phase signal is predominantly influenced by respiratory movements [24]. Furthermore, while breathing is typically slow and steady, heartbeats involve rapid muscle contractions. Consequently, the acceleration of breathing is generally less significant than that of the heartbeat. Therefore, the differentiator can effectively suppress the respiratory signal and enhance the heartbeat signal by calculating the acceleration (second derivative) of the mixed respiratory and heartbeat signals. Based on this principle, the following second-order differentiator [34] is employed, which exhibits robustness to noise:

$$A_p = \left(4 \times B_0 + \left(B_1 + B_{-1}\right) - 2 \times \left(B_2 + B_{-2}\right) - \left(B_3 + B_{-3}\right)\right) / \left(16 \times t^2\right) \quad (1)$$

where A_p is the second-order derivative at the current sample point, t is the sampling period, B_i is the sample point with sampling interval $(t \times i)$ from the current sampling point, $i = -3, -2, -1, 0, 1, 2, 3$.

Fig. 3A illustrates the radar phase signal, which is directly derived from the dataset. The phase of the radar phase signal refers to the displacement of human chest activity detected by the radar, mainly composed of respiratory movements and the heartbeat movements hidden within it. The RCG signal processed using formula (1) is presented in Fig. 3B. Fig. 3C and 3D in the red dotted boxes show the amplified RCG signal and its corresponding synchronously collected ECG signal, respectively. By comparing Fig. 3C and 3D, it can be seen that the waveform of RCG is significantly more complex than ECG. However, there is a similar rhythm between RCG and ECG. By comparing the two waveforms, it can be seen that there is a corresponding relationship between certain positions of RCG and fiducial points of ECG. Therefore, the location of ECG fiducial points can be directly inferred from RCG without collecting ECG signals.

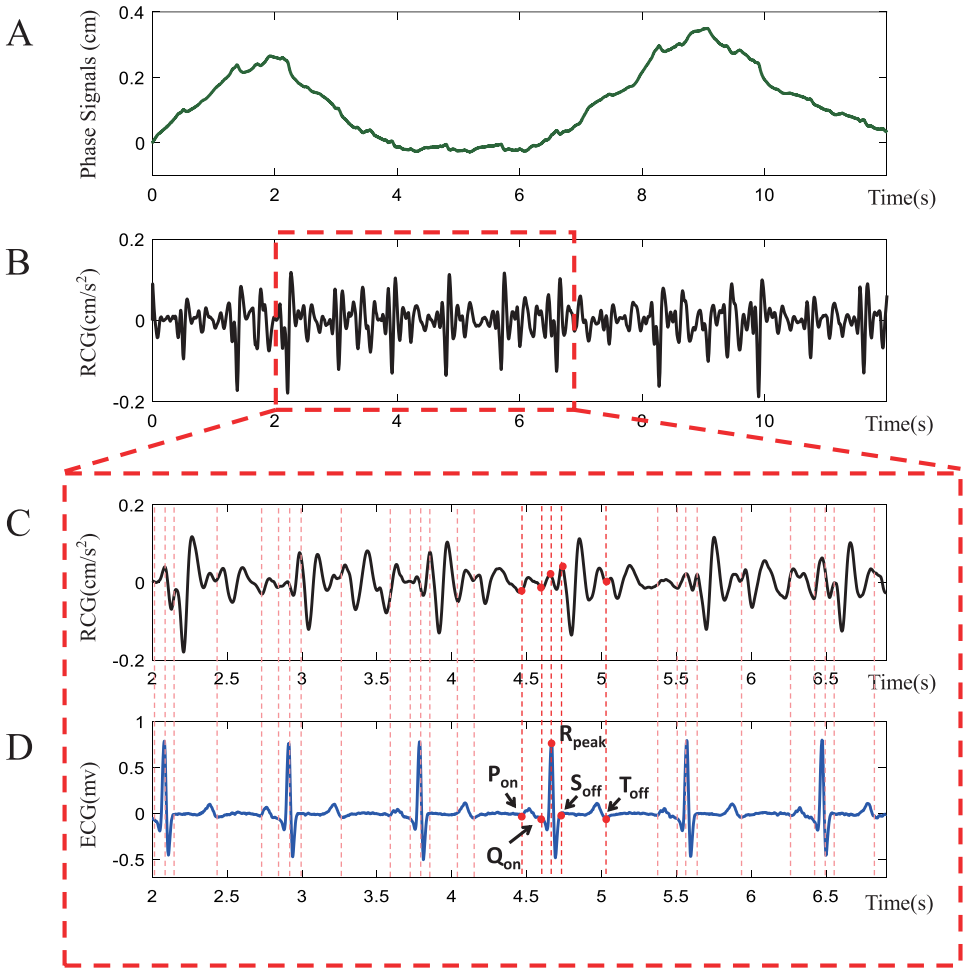

Fig. 3. The radar cardiogram (RCG) has a rhythmic correspondence with the reference electrocardiogram (A) The radar phase signal, (B) the RCG, (C) the amplified RCG signal, and (D) the synchronously collected ECG signal of the amplified RCG signal. P_{on} stands for onset of P-wave, Q_{on} for onset of QRS complex, R_{peak} for peak of R-wave, and S_{off} for offset of QRS complex.

2.1.2. ECG data preprocessing

Initially, a Butterworth low-pass filter was designed to eliminate myoelectric interference. Subsequently, zero-phase filtering was applied to remove baseline drift, followed by a 50 Hz notch filter to attenuate power frequency interference. Subsequently, automatic marking was performed on the ECG signal to generate the labels required for the deep learning network, followed by manual correction of the marking results. The selected ECG fiducial points include onset of P-wave (P_{on}), onset of QRS complex (Q_{on}), peak of R-wave (R_{peak}), offset of QRS complex (S_{off}), and offset of T-wave (T_{off}).

Utilizing the previous R_{peak} and T_{off} detection methods for ECG [29], along with the P-wave detection method [35], an automatic detection algorithm was developed for fiducial points in ECG. Additionally, a graphical user interface (GUI) was designed to facilitate manual calibration. In this study, two ECG analysis experts were invited to manually review and correct misidentified points produced by the algorithm, as well as to supplement any missing points. In cases where the experts' assessments were inconsistent, they collaboratively discussed the discrepancies to reach a final consensus.

The obtained fiducial points were used as reference points for generating semantic segmentation labels by segmenting ECG in order to meet the networks input. Semantic segmentation data for P_{on}, Q_{on}, R_{peak}, S_{off}, and T_{off} was obtained separately, denoted as P_{ss}, Q_{ss}, R_{ss}, S_{ss}, and T_{ss}. The specific segmentation annotation results are shown in Fig. 4A. Each semantic segmentation annotation contains only two values, 0 and 1, where the transition from 0 to 1 indicates the fiducial point position, as illustrated by the colored vertical lines in Fig. 4A. Furthermore, the number of sampling points in each semantic segmentation data is consistent with the number of sampling points in the ECG data.

2.2. Design of deep learning model and performance evaluation metrics

2.2.1. Design of deep learning model

In the network design section, a signal semantic segmentation learning network called DRSN-Unet is proposed, specifically designed for one-dimensional signal semantic segmentation. DRSN-Unet employs the U-shaped architecture of U-Net and utilizes skip connections, replacing the conventional downsampling convolution module with DRSN module [36]. The DRSN module is illustrated in Fig. 5A. Before the input data enters the DRSN module, initial features are incorporated into the output via an identity shortcut connection. The processing within the DRSN module begins with two convolutional layers, which serve to extract salient features from the input data. The output from these two convolutional layers undergoes an absolute value transformation followed by a global average pooling (GAP) operation. This GAP operation computes the mean of the features across each channel, effectively aggregating them and emphasizing the global statistical characteristics. This step reduces the number of parameters and compresses the dimensionality of the data to a single value per channel. Following this, two fully connected layers, along with a sigmoid activation function, are employed to generate threshold coefficients, denoted as α. The result of the GAP operation is then multiplied by α to derive an

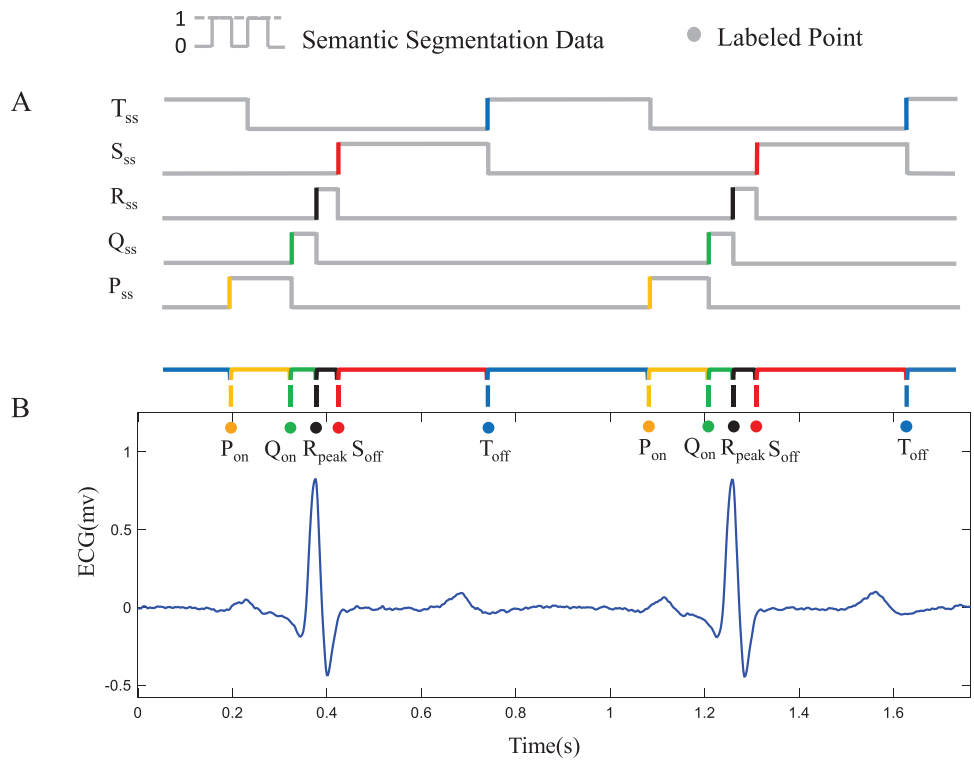

Fig. 4. Semantic segmentation data derived from labeled ECG fiducial points. (A) Semantic segmentation data of P_{on}, Q_{on}, R_{peak}, S_{off}, and T_{off}; (B) Reference ECG and labeled reference points. P_{on} stands for onset of P-wave, Q_{on} for onset of QRS complex, R_{peak} for peak of R-wave, and S_{off} for offset of QRS complex.

adaptive threshold (τ). τ is applied to soft thresholding, where the input of the soft threshold is obtained by the second convolutional layer to obtain the original features. Soft thresholding compares the amplitude of a signal against the predefined threshold τ to differentiate between meaningful signals and noise. If the amplitude of the signal is less than τ, it is set to zero. Conversely, if the amplitude is equal to or exceeds τ, the original signal is retained. This approach effectively suppresses noise while preserving the core characteristics of the signal, allowing for a focused analysis on relevant areas and minimizing the impact of extraneous noise. The formula for the soft threshold operation is as follows:

$$y = \begin{cases} x - \tau & x > \tau \\ 0 & -\tau \le x \le \tau \\ x + \tau & x < -\tau \end{cases} \tag{2}$$

where x represents the input features, y represents the output features, and τ denotes the adaptively adjustable threshold. The thresholding operation sets the features within the range of $[-\tau, \tau]$ to 0, providing a more flexible means of eliminating redundant information.

Fig. 5 illustrates the detailed structure of our deep learning network. Specifically, the architecture consists of a contracting path (left side) and an expansive path (right side). The module structure and the number of repetitions within the network are listed in Table 1. At the downsampling step, the input RCG signal undergoes 1D convolution, which reduces the data length while increasing the number of channels. Feature extraction is then performed through repeated 1D convolution and DRSN modules. In the upsampling step, deconvolution and 1D convolution operations are repeatedly applied to restore the data length and decrease the number of channels. Additionally, the network incorporates skip connections, which concatenate features of the same scale obtained from both downsampling and upsampling in the channel dimension. This approach effectively preserves and utilizes both deep and shallow information. In the final layer of the network, in accordance with the principles of semantic segmentation, 1D convolution was applied to expand the number of output channels from 1 to 5, corresponding to five types of semantic segmentation data. Furthermore, a post-processing function, designated as 'detect zero-to-one change function', was integrated into the network output stage. This function identifies the positions of zero-to-one transitions within the five semantic segmentation outputs, and these positions represent the locations corresponding to the fiducial points.

2.2.2. Design of performance evaluation metrics

Refer to the standards established by the Association for the Advancement of Medical Instrumentation (AAMI) [37], a tolerance range of 150 ms is used with the fiducial point as the central point. For instance, the true positive (TP), false positive (FP), false negative (FN), and true negative (TN) of the detected P_{on} using DRSN-Unet is calculated as follows (Fig. 6): if the detected P_{on} is located in the tolerance range, it is classified as TP for P_{on} detection; conversely, if the detected P_{on} is not located in this range, it is classified as FP; if there is no detected P_{on} in this range, it is classified as FN; the other undetected background points of RCG outside the range are all classified as TN. Similarly, TP, FP, FN, and TN of other fiducial points are calculated using this method. Notably, TN is too much larger than TP, FN, and FP, because the negative labeled samples (background samples) are far more than the positive labeled samples in the context of ECG fiducial point detection. Thus, this study evaluates the performance of different methods only using precision, recall, and $F_{1\text{-score}}$ metrics without accuracy and specificity. Recall indicates the algorithm's capability to detect fiducial points, while precision reflects the correctness of these detections. The $F_{1\text{-score}}$ serves as a crucial indicator that balances both recall and precision, thereby providing a comprehensive assessment of the algorithm's detection performance. For each type of fiducial point, the evaluation metrics used in this study are calculated according to the following formulas:

$$\text{Precision} = TP/(TP + FP) \tag{3}$$

$$\text{Recall} = TP/(TP + FN) \tag{4}$$

$$F_{1-score} = (2 \times \text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall}) \tag{5}$$

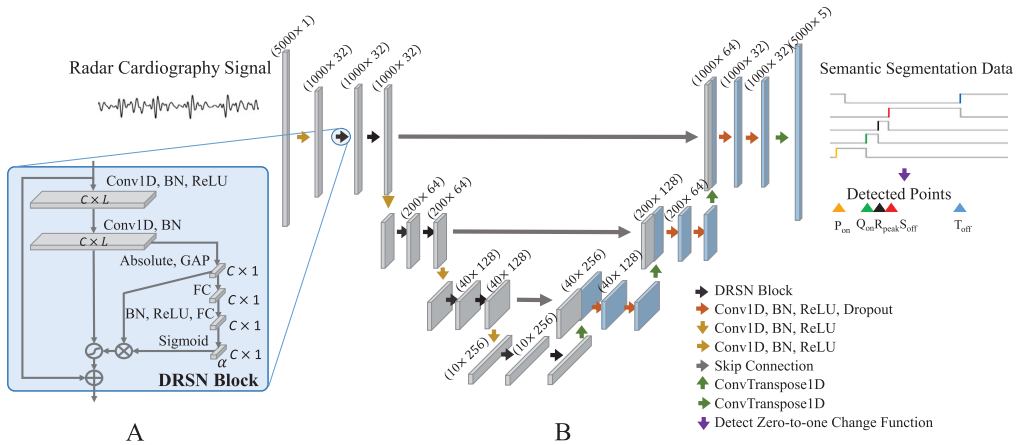

Fig. 5. The architecture of the proposed DRSN-Unet: (A) Deep residual shrinkage networks block (B) Overall architecture of the DRSN-Unet, where C and L are the indicators of the channel number and the feature-map length. Conv1D stands for one-dimensional convolution, BN for batch normalization, Trans-Conv1D for transposed one-dimensional convolution, ReLU for rectified linear unit, and FC for fully connected layer. The RCG is the input signal, which is processed by the network and finally outputs five semantic segmentation data. Then semantic segmentation data undergo a post-processing step to obtain the location information of the fiducial points.

Table 1
Architectural details of the proposed DRSN-Unet.

Block no.	Block Repeat umbers	Layer of block	Layer Repeat Numbers	Hyperparameter details
0	–	Input Layer	–	–
1	3	Conv1D +BN +ReLU	1	Kernel Size = 3, Stride = 5
		DRSN Block	2	Kernel Size = 3, Stride = 1
2	1	Conv1D +BN +ReLU	1	Kernel Size = 3, Stride = 4
		DRSN Block	2	Kernel Size = 3, Stride = 1
3	1	ConvTranspose1D	1	Kernel Size = 4, Stride = 4
		Conv1D + BN + ReLU +Dropout	2	Kernel Size = 3, Stride = 1, Dropout Rate = 0.4
4	2	ConvTranspose1D	1	Kernel Size = 5, Stride = 5
		Conv1D + BN + ReLU +Dropout	2	Kernel Size = 3, Stride = 1; Dropout Rate = 0.4
5	1	ConvTranspose1D	1	Kernel Size = 5, Stride = 5, Out Channel = 5

Conv1D stands for one-dimensional convolution, BN for batch normalization, ReLU for rectified linear unit, DRSN Block for deep residual shrinkage network module, and ConvTranspose1D for transposed one-dimensional convolution.

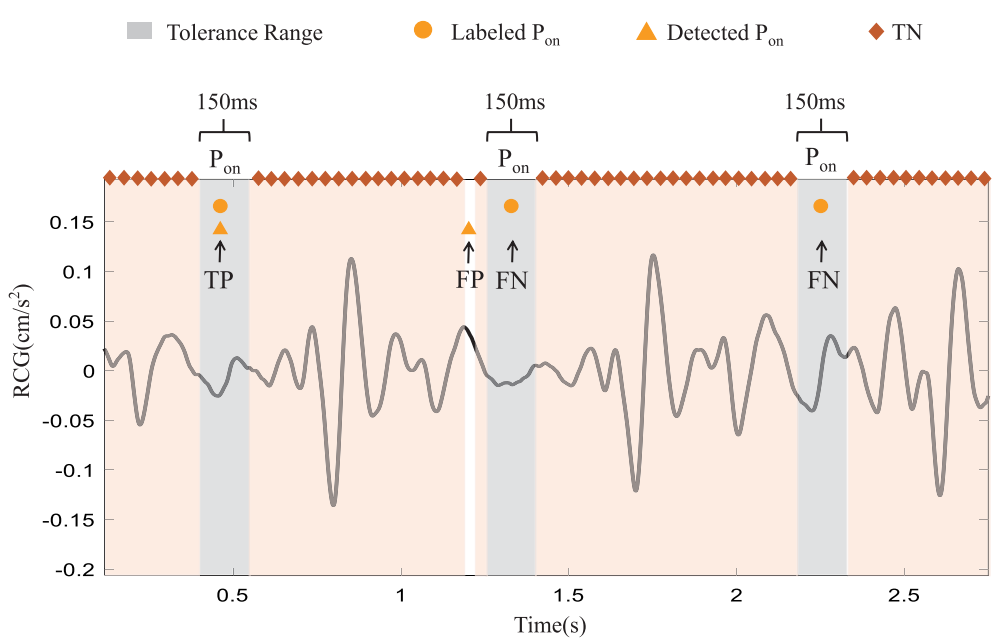

Fig. 6. Illustration of the true positive (TP), false positive (FP), false negative (FN), and true negatives (TN) for P_{on}. The gray area represents the tolerance range, circles represent label points, triangles represent detection points, and diamonds represent TN points.

The tolerance range in fiducial point detection directly impacts the precision of detection. A smaller tolerance will decrease detection precision. To verify the effect of tolerance range size on the detection, evaluation metrics are compared using different tolerance ranges of 150 ms, 100 ms, and 50 ms.

3. Experiment

3.1. Experimental data collection

The experimental data was sourced from a publicly available dataset developed by Professor Sven's team at the Hamburg Institute of Technology [33]. This dataset received approval from the Ethics Committee of Friedrich-Alexander-Universität Erlangen-Nürnberg (No. 85_15B). All experiments were conducted according to applicable guidelines and regulations, and informed consent was obtained from all subjects prior to participation in the human trials. A total of 30 healthy volunteers were involved, with data collected by physicians from Erlangen University Hospital following a prescribed protocol. The radar equipment utilized in the experiment was optimized based on the system detailed in Ref. [30]. Operating at a frequency of 24 GHz, the radar was equipped with a positioning laser to accurately target the subject's chest, positioned 40 centimeters away. The ECG signal acquisition was performed using the Task Force Monitor 3040i from CNSystems Medizintechnik GmbH, which allows for synchronous data collection alongside the radar system. This study selected the radar data and the synchronously collected ECG lead II data from all 30 subjects in the dataset for algorithm validation. All data employed for model training, validation, and testing were recorded under controlled conditions, specifically when subjects were in a supine position and engaged in uniform breathing. The total duration of data collection amounted to 19,048 s, resulting in an average data collection time of 635 s per subject.

Additionally, data from two subjects at a different data center were selected to validate the generalization and effectiveness of the model [24]. At this center, the AWR6843AOP radar from Texas Instruments, which operates at 60 GHz, was used. The seated posture during data collection may have introduced random motion interference. Therefore, data of two subjects with high quality signals were selected from this dataset for validation. And total duration of data collection amounted to 430 s, resulting in an average data collection time of 215 s per subject.

3.2. Reference fiducial points marking

Fig. 7 presents a GUI developed for the automatic localization of fiducial points within ECG signals, as well as for expert manual calibration. This GUI was created using Matlab R2021b software, facilitating both automated and manual processes to enhance the accuracy of ECG analysis.

3.3. Dataset partitioning

The dataset for model training and testing was divided in accordance with the principle of data exclusivity. Randomly selected data of 27 subjects from 30 subjects as the training and validation dataset, while data from the remaining 3 subjects constituted the testing dataset I. The model was trained on the training and validation dataset utilizing a 9-fold cross-validation approach. In each fold, the training and validation datasets were consistently divided in a 24:3 ratio of subjects. Furthermore, all data were segmented into 10-second time windows, resulting in a total of 1891 segments. Furthermore, data from 2 subjects were chosen from another data center, as cited in Ref. [24]. Each subject's data were segmented into 10-second intervals, yielding a total of 43 segments which constitute the testing dataset II. Due to the inconsistent recording durations of each subject, the number of data segments may vary in the training and validation datasets for each fold of cross-validation. Therefore, Table 2 presents the average number of data segments utilized during model training and validation, along with the number of segments used for testing.

3.4. System implementation and training

The workstation for training models was equipped with two NVIDIA Tesla V100 GPUs and a CPU with a maximum frequency of 3.40 GHz. The system had a total of 191 GB of RAM. The graphics card was an NVIDIA Quadro RTX 6000 with an operating memory of 24 GB. The operating system was 64-bit and Pytorch 1.9.1 was used for deep learning model training, validation, and testing.

The Root Mean Square Error Loss Function and the Adam optimizer were used to update the network parameters iteratively. The learning rate was set at 0.005, with a batch size of 512, and the training procedure consisted of 200 epochs. To mitigate the risk of overfitting and enhance training efficiency, all input data underwent normalization to the range [0, 1] through maximum-minimum scaling prior to being fed into the neural network.

4. Results and discussion

4.1. Nine-fold cross-validation results

To identify the optimal hyperparameter types and values, 9-fold cross-validation was conducted on the DRSN-Unet using the training and validation datasets. This approach not only facilitates hyperparameter tuning but also assesses the model's generalization performance. The 9-fold cross-validation results, reported as mean ± standard deviation (STD), were calculated for each of the five fiducial points, and the average detection performances of these five points were further evaluated (Table 3). Based on the analysis, the detection performance across the five fiducial points revealed that R_{peak} achieved the highest performance ($F_{1\text{-score}} = 0.960 \pm 0.046$). In contrast, P_{on} exhibited lower performance compared to other four points ($F_{1\text{-score}} = 0.933 \pm 0.045$). The average detection performances of these five points indicate that the DRSN-Unet demonstrates a robust capability for the precise detection of ECG fiducial points in radar cardiograms (average $F_{1\text{-score}} = 0.948 \pm 0.040$).

4.2. Comparison among different networks

After determining the types and quantities of model hyperparameters, the training dataset and validation dataset were combined into a new training dataset for further network parameter optimization, followed by model validation on the testing dataset. Five networks were selected for comparison: CNN-Transformer [22], MultiResLinkNet [23], Res-Unet [24], CNN-biLSTM [32], and 1D-Unet [31]. The testing dataset I was utilized to evaluate their performance. The results, presented in Table 4, were organized by ranking the evaluation metrics in the following order: P_{on}, Q_{on}, R_{peak}, S_{off}, T_{off}, and the overall mean value (calculated as the average of P_{on}, Q_{on}, R_{peak}, S_{off}, and T_{off}). The result of each fiducial point was arranged by $F_{1\text{-score}}$, from lowest to highest. As shown in Table 4, our designed network demonstrated superior detection performance for the five fiducial points, outperforming the other four networks (average precision = 0.986, average recall = 0.986, average $F_{1\text{-score}} = 0.986$). And for DRSN-Unet, it can be seen that R_{peak} and S_{off} have the best detection performance, while P_{on} has the worst detection performance, which may be related to the RCG waveform characteristics, but this relationship is not absolute, and may also be related to the selection of network model and the tolerance range.

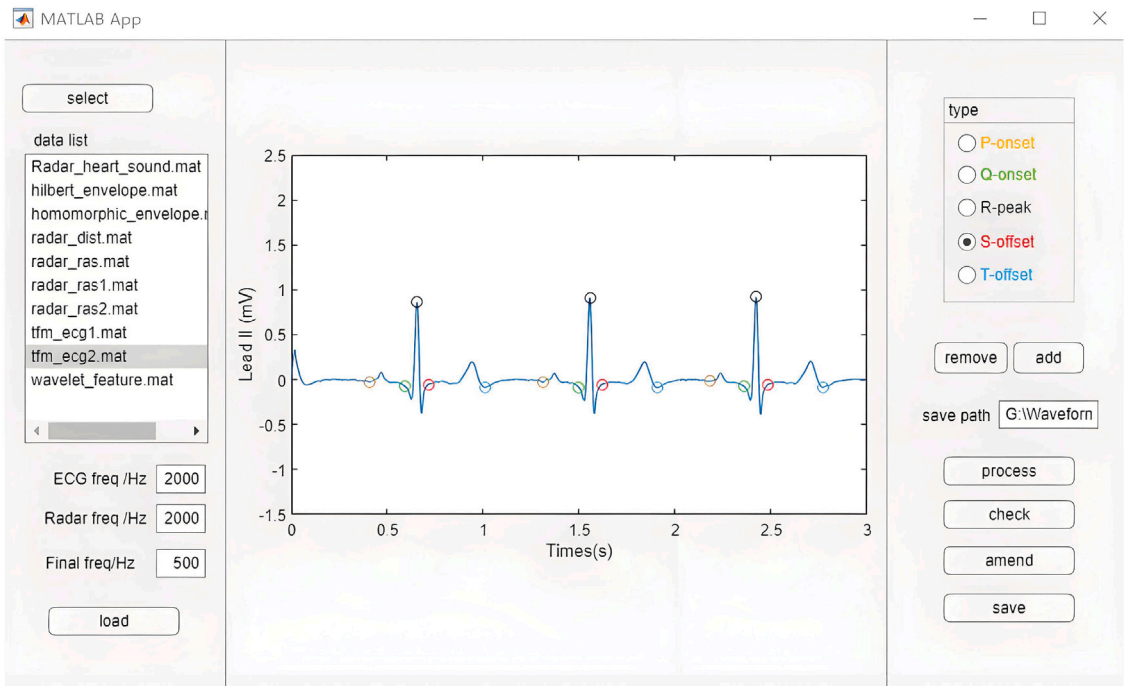

Fig. 7. Graphical user interface for reference fiducial points marking. The GUI features a dedicated section in the upper left corner for data selection. The lower left corner allows users to adjust the original and final sampling frequencies. By clicking the 'load' button, the system automatically detects fiducial points on the ECG and displays them in the central area of the interface. The upper right corner enables users to select the type of fiducial points. The lower part of the selection area provides options for performing various operations, such as adding or removing fiducial points, and managing data generation and storage paths.

Table 2
Details of dataset.

	Training dataset	Validation dataset	Testing dataset I	Testing dataset II	Total
Number of Segments	1517	191	183	43	1934

Table 3
The 9-fold cross-validation and the average detection performances of five fiducial points using DRSN-Unet.

Type	Metrics		
	Precision	Recall	$F_{1\text{-score}}$
P_{on}	0.933 ± 0.045	0.933 ± 0.047	0.933 ± 0.045
Q_{on}	0.962 ± 0.024	0.948 ± 0.043	0.955 ± 0.032
R_{peak}	0.967 ± 0.029	0.953 ± 0.065	0.960 ± 0.046
S_{off}	0.930 ± 0.040	0.944 ± 0.034	0.937 ± 0.037
T_{off}	0.951 ± 0.045	0.965 ± 0.037	0.958 ± 0.040
Average	0.949 ± 0.037	0.948 ± 0.045	0.948 ± 0.040

P_{on} stands for onset of P-wave, Q_{on} for onset of QRS complex, R_{peak} for peak of R-wave, S_{off} for offset of QRS complex, and Average for the average detection performances of five fiducial points.

A comparison was also made of the computational complexity and parameters of each network, as summarized in Table 5. MultiResLinkNet and CNN-Transformer are both networks used for ECG reconstruction tasks. Their structures were more complex compared to DRSN-Unet (Δparameters = 7.084 and 1.149 M, respectively), with longer running times than DRSN-Unet (ΔFLOPs = 45.563 and 15.986 GFLOPs, respectively) and poorer performance than DRSN-Unet (Δaverage $F_{1\text{-score}}$ = −0.148 and −0.141, respectively) in fiducial point detection tasks. Res-Unet is also a network used for ECG reconstruction and demonstrates good application capabilities for fiducial point detection tasks. However, its average detection performance was lower than that of DRSN-Unet (Δaverage $F_{1\text{-score}}$ = −0.027). CNN-biLSTM and 1D-Unet are networks designed for one-dimensional data segmentation and fiducial point localization, both of which feature have relatively simple structures. Among them, CNN-biLSTM employed a multi-layer convolutional structure, resulting in high computational complexity (FLOPs = 6.014 GFLOPs) but a small number of network parameters (Parameters = 1.196 M). However, both CNN-biLSTM and 1D-Unet underperform compared to DRSN-Unet (Δaverage $F_{1\text{-score}}$ = −0.037 and −0.058, respectively) in the task of locating ECG fiducial points based on millimeter wave radar. DRSN-Unet utilized an adaptive soft thresholding method to eliminate redundant information, thereby effectively isolating and identifying useful feature information. Overall, these results demonstrated that DRSN-Unet outperformed the other four models while maintaining a low computational load and parameter count, underscoring its efficiency in fiducial point detection.

4.3. Comparison among different center datasets

To verify the generalization ability, reliability, and fairness of deep learning models and ensure their effectiveness across diverse data and environments, testing dataset II was used to validate the performance of the trained network model. The results are shown in Table 6. It can be observed that, compared to testing dataset I, the precision, recall and $F_{1\text{-score}}$ of testing dataset II are all reduced (Δaverage precision = −0.117, Δaverage recall = −0.175, and Δaverage $F_{1\text{-score}}$ = −0.154).

The first reason for this decline is that different centers used different radars to collect data, which introduced subtle differences in the captured signals. Second, the data in testing dataset II was collected while the subjects were in sitting posture, with random body motion interference. Although the body motion interference suppression method in [24] was used to decrease partial interference, there is

Table 4
Comparison of the detection performances among different networks.

Type	Model	Metrics		
		Precision	Recall	$F_{1\text{-score}}$
P_{on}	MultiResLinkNet [23]	0.741	0.863	0.797
	CNN-Transformer [22]	0.957	0.765	0.850
	1D-Unet [31]	0.858	0.879	0.868
	Res-Unet [24]	0.906	0.954	0.930
	CNN-biLSTM [32]	0.966	0.956	0.961
	DRSN-Unet	**0.975**	**0.975**	**0.975**
Q_{on}	CNN-Transformer [22]	0.970	0.683	0.802
	MultiResLinkNet [23]	0.876	0.945	0.909
	1D-Unet [31]	0.941	0.935	0.938
	Res-Unet [24]	0.970	0.979	0.974
	CNN-biLSTM [32]	0.979	0.972	0.975
	DRSN-Unet	**0.986**	**0.984**	**0.985**
R_{peak}	CNN-Transformer [23]	0.968	0.866	0.914
	MultiResLinkNet [22]	0.924	0.982	0.952
	1D-Unet [31]	0.966	0.947	0.957
	Res-Unet [24]	0.946	0.980	0.963
	CNN-biLSTM [32]	0.970	0.971	0.970
	DRSN-Unet	**0.996**	**0.991**	**0.993**
S_{off}	MultiResLinkNet [22]	0.738	0.964	0.836
	CNN-Transformer [23]	0.857	0.914	0.884
	1D-Unet [31]	0.915	0.960	0.937
	Res-Unet [24]	0.946	0.980	0.963
	CNN-biLSTM [32]	0.960	0.984	0.972
	DRSN-Unet	**0.992**	**0.996**	**0.994**
T_{off}	MultiResLinkNet [23]	0.619	0.789	0.694
	CNN-Transformer [22]	0.702	0.862	0.774
	CNN-biLSTM [32]	0.855	0.878	0.867
	1D-Unet [31]	0.939	0.939	0.939
	Res-Unet [24]	0.950	0.982	0.966
	DRSN-Unet	**0.982**	**0.985**	**0.983**
Average	MultiResLinkNet [23]	0.780	0.910	0.838
	CNN-Transformer [22]	0.891	0.818	0.845
	1D-Unet [31]	0.924	0.932	0.928
	CNN-biLSTM [32]	0.946	0.952	0.949
	Res-Unet [24]	0.944	0.975	0.959
	DRSN-Unet	**0.986**	**0.986**	**0.986**

P_{on} stands for onset of P-wave, Q_{on} for onset of QRS complex, R_{peak} for peak of R-wave, S_{off} for offset of QRS complex, and average for the overall mean metrics value of five fiducial point detections.

Table 5
Computational complexity and model sizes of different networks.

Model	Metrics	
	FLOPs (GFLOPs)	Network parameters (M)
MultiResLinkNet [23]	45.649	9.126
CNN-Transformer [22]	16.072	3.191
CNN-biLSTM [32]	6.014	1.197
Res-Unet [24]	0.084	1.332
1D-Unet [31]	0.083	1.289
DRSN-Unet	0.086	2.042

GFLOPs stands for Giga Floating Point Operations Per Second, and M for Mega.

Table 6
Comparison of the detection performances among different testing datasets.

Type	Testing dataset	Metrics		
		Precision	Recall	$F_{1\text{-score}}$
P_{on}	I	0.975	0.975	0.975
	II	**0.863**	**0.843**	**0.853**
Q_{on}	I	0.986	0.984	0.985
	II	**0.912**	**0.781**	**0.814**
R_{peak}	I	0.996	0.991	0.993
	II	**0.928**	**0.755**	**0.833**
S_{off}	I	0.992	0.996	0.994
	II	**0.859**	**0.866**	**0.862**
T_{off}	I	0.982	0.985	0.983
	II	**0.784**	**0.811**	**0.797**
Average	I	0.986	0.986	0.986
	II	**0.869**	**0.811**	**0.832**

still some interference that further leads to the reduction in detection performance. Nevertheless, overall, DRSN-Unet achieved an approximate score of 0.8 on data from another center, demonstrating its continued effectiveness in fiducial point detection despite variations in data collection conditions.

4.4. Comparison among different tolerance ranges

When the tolerance range was narrowed to 100 ms and 50 ms, the performance of ECG fiducial point detection using the DRSN-Unet under these varying conditions was evaluated with testing dataset I. In Table 7, the performance metrics across different tolerance ranges were summarized, and the maximum difference in performance indices was calculated as the tolerance decreased from 150 ms to 50 ms. As indicated in Table 7, the designed model achieved an average $F_{1\text{-score}}$ of 0.986 within a 150 ms tolerance range. However, when the tolerance was reduced to 100 ms, the average $F_{1\text{-score}}$ decreases to 0.967. Further reducing the tolerance to 50 ms resulted in a significant decline across all evaluation metrics, with the average $F_{1\text{-score}}$ dropping to 0.849-a reduction of 0.137. The most pronounced impact was observed at Q_{on}, where the $F_{1\text{-score}}$ decreased by 0.250. Conversely, the decrease in $F_{1\text{-score}}$ for the R_{peak} was minimal, at only 0.029. As the tolerance range is narrowed, the evaluation conditions become more stringent. The results showed that the points which originally fell within the tolerance range may now fall outside it, leading to a decrease in TP values, while FP and FN values increased, ultimately causing a decline in all three evaluation metrics.

Table 7
Comparison of the detection performances among different tolerance ranges using DRSN-Unet.

Type	Tolerance range	Metrics		
		Precision	Recall	$F_{1\text{-score}}$
P_{on}	150 ms	0.974	0.975	0.975
	100 ms	0.956	0.957	0.957
	50 ms	0.790	0.791	0.790
	Maximum Difference	**0.184**	**0.184**	**0.185**
Q_{on}	150 ms	0.986	0.984	0.985
	100 ms	0.942	0.941	0.942
	50 ms	0.736	0.735	0.736
	Maximum Difference	**0.250**	**0.249**	**0.249**
R_{peak}	150 ms	0.996	0.991	0.993
	100 ms	0.995	0.970	0.992
	50 ms	0.966	0.962	0.964
	Maximum Difference	**0.030**	**0.029**	**0.029**
S_{off}	150 ms	0.992	0.996	0.994
	100 ms	0.986	0.990	0.988
	50 ms	0.913	0.920	0.916
	Maximum Difference	**0.079**	**0.076**	**0.078**
T_{off}	150 ms	0.982	0.985	0.983
	100 ms	0.957	0.960	0.959
	50 ms	0.837	0.842	0.839
	Maximum Difference	**0.145**	**0.143**	**0.144**
Average	150 ms	0.986	0.986	0.986
	100 ms	0.967	0.967	0.967
	50 ms	0.848	0.850	0.849
	Maximum Difference	**0.138**	**0.136**	**0.137**

P_{on} stands for onset of P-wave, Q_{on} for onset of QRS complex, R_{peak} for peak of R-wave, S_{off} for offset of QRS complex, Average for the overall mean metrics value of five fiducial point detections, and Maximum Difference for the maximum difference in performance metrics as the tolerance decreases from 150 ms to 50 ms.

5. Conclusion and future work

A novel non-contact ECG fiducial point detection method utilizing millimeter-wave radar is proposed. A semantic segmentation network model, termed DRSN-Unet, has been designed specifically for fiducial point detection from RCG. This model integrates the advantages of the U-net architecture and deep residual shrinkage networks, thereby effectively extracting features from one-dimensional signals. The DRSN-Unet model is capable of detecting five types of fiducial points in ECG, including the onset of the P-wave and QRS complex, the peak of the R-wave, the offset of the QRS complex and T-wave. When compared to the other five existing excellent network models, our DRSN-Unet demonstrates superior performance across three key metrics, achieving an average precision of 0.986, an average recall of 0.986, and an average $F_{1\text{-score}}$ of 0.986. Additionally, this model has a compact architecture, with a total parameter size of only 2.042 M and a computational complexity of 0.086 GFLOPs. Furthermore, the impact of tolerance range size on the fiducial point detection performance metrics was examined, finding that smaller tolerance ranges correlate with decreased detection performance.

However, when validating with data from another center, it was observed that the $F_{1\text{-score}}$ of the model decreased due to the slightly different data distribution from the other center and the interference from random body motion. In future research, random body motion interference would be suppressed by designing adaptive filtering algorithms, utilizing multi-sensor synchronous measurements, and eliminating noisy segments. Moreover, the model's generalization will be improved by suppressing RCG interference, increasing multi-center training data, and designing a better model architecture. In addition, the data utilized in this study were collected exclusively from healthy subjects, resulting in a lack of information on arrhythmia patients and insufficient discussion of fiducial point positioning under arrhythmic conditions. And the study focused solely on fiducial point detection without delving into ECG feature extraction analysis. Therefore, future research will involve expanding the dataset by collecting radar and simultaneous ECG data from cardiac arrhythmia patients. The same architecture will be employed for fiducial point detection to assist in clinical diagnosis. From a technological perspective, existing chip radar technology has achieved centimeter-level sizes. Due to its small size and mature technology, the cost of mass production is low. In the industrial sector, the frequency range commonly used by millimeter-wave radar is primarily concentrated at 24 GHz, 60 GHz, 77 GHz, and so on, with little electromagnetic interference from medical devices. Moreover, the adoption of radar technology in medical applications necessitates the establishment of more comprehensive laws and regulations to ensure its effective and safe use.

CRediT authorship contribution statement

Binyue Cao: Data curation, Software, Visualization, Writing – original draft, Conceptualization. **Mi He:** Funding acquisition, Writing – review & editing, Conceptualization, Methodology. **Meiyun Zhao:** Software, Formal analysis, Writing – original draft. **Qinwen Ping:** Conceptualization, Formal analysis, Methodology. **Chang He:** Visualization, Validation, Formal analysis. **Xiangyu Zhou:** Software, Validation, Resources. **Yushun Gong:** Data curation, Supervision.

Informed consent

Not applicable.

Declaration of competing interest

No potential conflict of interest was reported by the authors. The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The authors do not have permission to share data.

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